

ITFS TASK 1

DEVOPS PRACTICAL MISSION

GitHub Repository: <https://www.github.com/adithyansk-2002/AWS-Ansible-Automation>

1. Summary

This project demonstrates the deployment and automation of an Ansible environment on AWS EC2 across two AWS regions. Mission 1 focuses on building the environment in Mumbai, while Mission 2 recreates the same environment in Hyderabad by cloning the GitHub repository and reusing the existing playbooks.

All instance access was performed through AWS EC2 Instance Connect. SSH connectivity between the control node and managed nodes was established by manually sharing the public SSH key from the control node to each client instance.

2. Approach and Methodology

2.1 Access and Connectivity

- Launched EC2 instances in AWS Console (Mumbai — Mission 1; Hyderabad — Mission 2).
- Connected via EC2 Instance Connect from the AWS Console.
- Created devops (Mission 1) or clone (Mission 2) user on all machines.
- Generated SSH keys on the control node and manually copied the public key to `authorized_keys` on clients.
- Verified SSH from control node to both clients before running Ansible.

2.2 Automation Workflow

- Installed Docker, Python 3, pip, Git, Ansible, and ansible-navigator on the control node.
- Created project directory: `/home/devops/ansible` (Mission 1) or `/home/clone/ansible` (Mission 2).
- Authored inventory, `ansible.cfg`, playbooks, and `roles/myrole`.
- Configured `ansible-navigator.yml`.
- Executed all playbooks using `#ansible-navigator run <playbook>.yml -m stdout`.
- Verified results via SSH, curl, and browser.
- Pushed contents to GitHub.

3. Mission 1 - AWS Infrastructure Deployment (Mumbai - ap-south-1)

Instances: mumbai-control, mumbai-client1 (dev), mumbai-client2 (test)

User: devops

Path: /home/devops/ansible

Task 1: Create AWS Infrastructure

Objective: Create three EC2 instances in Mumbai, create the devops user, and configure SSH key-based authentication.

- Launched three t3.micro instances in ap-south-1a with prescribed naming.
- Created devops user on all instances via EC2 Instance Connect.
- Shared SSH public keys manually between control node and clients.
- Verified passwordless SSH from mumbai-control to both clients.

Step 1: Run 3 instances in EC2 Console in Mumbai (mumbai-control, mumbai-client1, mumbai-client2)

The screenshot displays the AWS Management Console's EC2 Instances page. The left sidebar shows the navigation menu with 'Instances' selected. The main content area shows a list of three instances: 'mumbai-client2', 'mumbai-control', and 'mumbai-client1'. All instances are in the 'Running' state and are t3.micro instances located in the ap-south-1a availability zone. The console also shows a search bar, filter buttons, and a 'Launch instances' button.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
mumbai-client2	i-03a74275e693e94b8	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-127-215-162.ap...	1
mumbai-control	i-0ac96637a9cf101be	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-201-68-56.ap-s...	1
mumbai-client1	i-07a8d899f2b9ad292	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-52-66-203-197.ap...	5

Step 2: Verify SSH connectivity from control node to mumbai-client1 and mumbai-client2 after sharing public keys to both instances

```
[devops@ip-172-31-38-67 ~]$ ssh devops@52.66.203.197
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[devops@ip-172-31-32-198 ~]$ exit
logout
Connection to 52.66.203.197 closed.
[devops@ip-172-31-38-67 ~]$
[devops@ip-172-31-38-67 ~]$ ssh devops@13.127.215.162
The authenticity of host '13.127.215.162 (13.127.215.162)' can't be established.
ED25519 key fingerprint is SHA256:tr5iqqrsukhAZcEgNX3/P6EKtchw5AY7Qv4qtceAd00.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.127.215.162' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[devops@ip-172-31-40-62 ~]$ exit
logout
```

Task 2: Install and Configure Ansible Environment

Objective: Configure the Ansible inventory and ansible.cfg file, then verify connectivity with all managed nodes.

- Installed and enabled Docker on mumbai-control.
- Installed Python 3, pip, ansible-navigator, and ansible-builder.
- Created inventory with mumbai-client1 in dev group and mumbai-client2 in test group.
- Created ansible.cfg pointing to /home/devops/ansible/inventory.
- Verified connectivity with `ansible all -m ping` — both hosts returned pong.

Step 1: Install Docker on mumbai-control via dnf

```
[devops@ip-172-31-38-67 ansible]$ sudo dnf install -y docker
Last metadata expiration check: 0:00:14 ago on Tue Jun 23 07:00:50 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
docker	x86_64	25.0.16-1.amzn2023.0.2	amazonlinux	46 M
Installing dependencies:				
container-selinux	noarch	4:2.245.0-1.amzn2023	amazonlinux	58 k
containerd	x86_64	2.2.4-1.amzn2023.0.3	amazonlinux	24 M
iptables-libs	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	401 k
iptables-nft	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	183 k
libcgroup	x86_64	3.0-1.amzn2023.0.1	amazonlinux	75 k
libnetfilter_conntrack	x86_64	1.0.8-2.amzn2023.0.2	amazonlinux	58 k
libnftnl	x86_64	1.0.1-19.amzn2023.0.2	amazonlinux	30 k
libnftnl	x86_64	1.2.2-2.amzn2023.0.2	amazonlinux	84 k
pigz	x86_64	2.5-1.amzn2023.0.4	amazonlinux	83 k
runc	x86_64	1.3.5-1.amzn2023.0.2	amazonlinux	3.9 M

```
Transaction Summary
Install 11 Packages

Total download size: 76 M
Installed size: 284 M
Downloading Packages:
(1/11): container-selinux-2.245.0-1.amzn2023.noarch.rpm      1.5 MB/s | 58 kB    00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm      14 MB/s | 401 kB   00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm        5.9 MB/s | 183 kB   00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm            2.7 MB/s | 75 kB    00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm 2.0 MB/s | 58 kB    00:00
(6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm           1.0 MB/s | 30 kB    00:00
```

Step 2: Enable and start Docker service

```
[devops@ip-172-31-38-67 ansible]$ sudo systemctl enable --now docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[devops@ip-172-31-38-67 ansible]$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-06-23 07:04:58 UTC; 4s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Process: 29952 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
    Process: 29953 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Main PID: 29954 (dockerd)
      Tasks: 9
     Memory: 27.3M
        CPU: 362ms
    CGroup: /system.slice/docker.service
            └─29954 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nfile=32768:65536

Jun 23 07:04:56 ip-172-31-38-67.ap-south-1.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Jun 23 07:04:56 ip-172-31-38-67.ap-south-1.compute.internal dockerd[29954]: time="2026-06-23T07:04:56.919029077Z" level=info msg="Starting up"
Jun 23 07:04:56 ip-172-31-38-67.ap-south-1.compute.internal dockerd[29954]: time="2026-06-23T07:04:56.977288408Z" level=info msg="[graphdriver] using prior storage driver: overlay2"
Jun 23 07:04:56 ip-172-31-38-67.ap-south-1.compute.internal dockerd[29954]: time="2026-06-23T07:04:56.977536589Z" level=info msg="Loading containers: start."
Jun 23 07:04:57 ip-172-31-38-67.ap-south-1.compute.internal dockerd[29954]: time="2026-06-23T07:04:57.791207355Z" level=info msg="Default bridge (docker0) is already up. IP address: 172.17.0.1"
Jun 23 07:04:57 ip-172-31-38-67.ap-south-1.compute.internal dockerd[29954]: time="2026-06-23T07:04:57.976794267Z" level=info msg="Loading containers: done."
Jun 23 07:04:58 ip-172-31-38-67.ap-south-1.compute.internal dockerd[29954]: time="2026-06-23T07:04:58.001249601Z" level=info msg="Docker daemon" commit=6fdf0a6_n=25.0.16 builddate=2025-09-16T13:08:17Z
Jun 23 07:04:58 ip-172-31-38-67.ap-south-1.compute.internal dockerd[29954]: time="2026-06-23T07:04:58.001324361Z" level=info msg="Daemon has completed initialization"
Jun 23 07:04:58 ip-172-31-38-67.ap-south-1.compute.internal dockerd[29954]: time="2026-06-23T07:04:58.012267204Z" level=info msg="API listen on /run/docker.sock"
Jun 23 07:04:58 ip-172-31-38-67.ap-south-1.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
```

Step 3: Install Python 3 and pip

```
[devops@ip-172-31-38-67 ansible]$ sudo dnf install -y python3 python3-pip
Last metadata expiration check: 0:08:50 ago on Tue Jun 23 07:00:50 2026.
Package python3-3.9.25-1.amzn2023.0.6.x86_64 is already installed.
Dependencies resolved.

=====
Package                                Architecture      Version                                Repository        Size
=====
Installing:
python3-pip                            noarch            21.3.1-2.amzn2023.0.20                amazonlinux       1.8 M
Installing weak dependencies:
libxcrypt-compat                       x86_64            4.4.33-7.amzn2023                      amazonlinux       92 k
=====
Transaction Summary
-----
Install 2 Packages

Total download size: 1.9 M
Installed size: 11 M
Downloading Packages:
(1/2): libxcrypt-compat-4.4.33-7.amzn2023.x86_64.rpm                2.3 MB/s | 92 kB  00:00
(2/2): python3-pip-21.3.1-2.amzn2023.0.20.noarch.rpm               30 MB/s | 1.8 MB  00:00
-----
Total
-----
19 MB/s | 1.9 MB  00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :
  Installing     : libxcrypt-compat-4.4.33-7.amzn2023.x86_64        1/1
  Installing     : python3-pip-21.3.1-2.amzn2023.0.20.noarch        1/2
  Installing     : python3-pip-21.3.1-2.amzn2023.0.20.noarch        2/2
  Running scriptlet: python3-pip-21.3.1-2.amzn2023.0.20.noarch      2/2
  Verifying      : libxcrypt-compat-4.4.33-7.amzn2023.x86_64        1/2
```

Step 4: Install ansible-navigator and ansible-builder

```
[devops@ip-172-31-38-67 ansible]$ python3 -m pip install --user ansible-navigator ansible-builder
Collecting ansible-navigator
  Downloading ansible_navigator-24.2.0-py3-none-any.whl (297 kB)
    |#####| 297 kB 16.9 MB/s
Collecting ansible-builder
  Downloading ansible_builder-3.1.1-py3-none-any.whl (46 kB)
    |#####| 46 kB 7.7 MB/s
Collecting tzdata
  Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
    |#####| 349 kB 63.7 MB/s
Collecting ansible-runner<3,>=2.3.1
  Downloading ansible_runner-2.4.3-py3-none-any.whl (80 kB)
    |#####| 80 kB 15.1 MB/s
Requirement already satisfied: pyyaml in /usr/lib64/python3.9/site-packages (from ansible-navigator) (5.4.1)
Requirement already satisfied: jsonschema in /usr/lib/python3.9/site-packages (from ansible-navigator) (3.2.0)
Collecting ansible-lint>=6.19.0
  Downloading ansible_lint-6.22.2-py3-none-any.whl (297 kB)
    |#####| 297 kB 63.2 MB/s
Requirement already satisfied: jinja2 in /usr/lib/python3.9/site-packages (from ansible-navigator) (2.11.3)
Collecting importlib-metadata
  Downloading importlib_metadata-8.7.1-py3-none-any.whl (27 kB)
Collecting onigurumacffi<2,>=1.1.0
  Downloading onigurumacffi-1.4.1-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (603 kB)
    |#####| 603 kB 66.2 MB/s
Collecting bindep
  Downloading bindep-2.14.0-py3-none-any.whl (35 kB)
Collecting packaging
  Downloading packaging-26.2-py3-none-any.whl (100 kB)
    |#####| 100 kB 17.4 MB/s
Collecting ruamel.yaml>=0.18.5
  Downloading ruamel_yaml-0.19.1-py3-none-any.whl (118 kB)
    |#####| 118 kB 65.9 MB/s
```

Step 5: Version verification — ansible-navigator, ansible, and Docker

```
[devops@ip-172-31-38-67 ansible]$ source ~/.bashrc
[devops@ip-172-31-38-67 ansible]$ ansible-navigator --version
ansible-navigator 24.2.0
[devops@ip-172-31-38-67 ansible]$ ansible --version
ansible [core 2.15.13]
  config file = None
  configured module search path = ['/home/devops/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /home/devops/.local/lib/python3.9/site-packages/ansible
  ansible collection location = /home/devops/.ansible/collections:/usr/share/ansible/collections
  executable location = /home/devops/.local/bin/ansible
  python version = 3.9.25 (main, May 25 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
[devops@ip-172-31-38-67 ansible]$ docker --version
Docker version 25.0.14, build 0bab007
[devops@ip-172-31-38-67 ansible]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

Step 6: Inventory file configuration (dev and test groups)

```
[devops@ip-172-31-38-67 ansible]$ vim inventory
[devops@ip-172-31-38-67 ansible]$ cat inventory
[dev]
mumbai-client1 ansible_host=3.108.62.44

[test]
mumbai-client2 ansible_host=13.206.195.223

[all:vars]
ansible_user=devops
ansible_python_interpreter=/usr/bin/python3
```

Step 7: ansible.cfg configuration

```
[devops@ip-172-31-38-67 ansible]$ vim ansible.cfg
[devops@ip-172-31-38-67 ansible]$ cat ansible.cfg
[defaults]
inventory = /home/devops/ansible/inventory
host_key_checking = False
retry_files_enabled = False
remote_user = devops
```

Step 8: Successful ansible all -m ping connectivity test

```
[devops@ip-172-31-38-67 ansible]$ ansible all -m ping
mumbai-client1 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
mumbai-client2 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

Task 3: Package Management Automation

Objective: Create and execute packages.yml — MariaDB, PHP on dev and test; package updates and Development Tools on dev only.

Step 1: Create packages.yml playbook

```
[devops@ip-172-31-38-67 ansible]$ vim packages.yml
[devops@ip-172-31-38-67 ansible]$ cat packages.yml
---
- name: Install common packages on dev and test
  hosts: dev:test
  become: true
  tasks:
    - name: Install MariaDB package and PHP
      dnf:
        name:
          - mariadb105
          - php
        state: present

- name: Update all packages and install Development Tools on dev
  hosts: dev
  become: true
  tasks:
    - name: Update all packages on dev
      dnf:
        name: "*"
        state: latest

    - name: Install Development Tools group on dev
      dnf:
        name: "@Development Tools"
        state: present
```

Step 2: ansible-navigator.yml execution environment configuration

```
[devops@ip-172-31-38-67 ansible]$ vim ansible-navigator.yml
[devops@ip-172-31-38-67 ansible]$ cat ansible-navigator.yml
---
ansible-navigator:
  execution-environment:
    enabled: true
    image: ghcr.io/ansible/creator-ee:v0.22.0
    pull:
      policy: missing
  ansible:
    config:
      path: /home/devops/ansible/ansible.cfg
```

Step 3: Execute packages.yml using ansible-navigator

```
[devops@ip-172-31-38-67 ansible]$ ansible-navigator run packages.yml -m stdout

PLAY [Install common packages on dev and test] *****

TASK [Gathering Facts] *****
ok: [mumbai-client2]
ok: [mumbai-client1]

TASK [Install MariaDB package and PHP] *****
changed: [mumbai-client1] => (item=mariadb)
changed: [mumbai-client1] => (item=php)
changed: [mumbai-client2] => (item=mariadb)
changed: [mumbai-client2] => (item=php)

PLAY [Update all packages and install Development Tools on dev] *****

TASK [Gathering Facts] *****
ok: [mumbai-client2]

TASK [Update all packages on dev] *****
ok: [mumbai-client2]

TASK [Install Development Tools group on dev] *****
changed: [mumbai-client2]

PLAY RECAP *****
mumbai-client1      : ok=2      changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
mumbai-client2      : ok=5      changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Task 4: Create and Use Roles

Objective: Create roles/myrole with Jinja2 template; configure Apache on test group only.

Template output: Welcome to <full hostname> on <IP address>

Step 1: Configure roles/myrole/tasks/main.yml — Apache and template tasks

```
[devops@ip-172-31-38-67 ansible]$ vim roles/myrole/tasks/main.yml
[devops@ip-172-31-38-67 ansible]$ cat roles/myrole/tasks/main.yml
---
- name: Install Apache web server
  dnf:
    name: httpd
    state: present

- name: Deploy index page from template
  template:
    src: index.j2
    dest: /var/www/html/index.html
    mode: "0644"

- name: Start and enable Apache
  service:
    name: httpd
    state: started
    enabled: true
```

Step 2: Configure myrole.yml playbook targeting test group

```
[devops@ip-172-31-38-67 ansible]$ vim myrole.yml
[devops@ip-172-31-38-67 ansible]$ cat myrole.yml
---
- name: Apply myrole to test group
  hosts: test
  become: true
  roles:
    - myrole
```

Step 3: Successful execution of myrole.yml on mumbai-client2

```
[devops@ip-172-31-38-67 ansible]$ ansible-navigator run myrole.yml -m stdout

PLAY [Apply myrole to test group] *****

TASK [Gathering Facts] *****
ok: [mumbai-client2]

TASK [myrole : Install Apache web server] *****
ok: [mumbai-client2]

TASK [myrole : Start and enable Apache] *****
changed: [mumbai-client2]

TASK [myrole : Create test server homepage] *****
changed: [mumbai-client2]

PLAY RECAP *****
mumbai-client2      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Step 4: Browser verification — test web page showing hostname and IP address



Task 5: Content Modification in /etc/issue

Objective: Replace /etc/issue content — development (dev) and test (test).

Step 1: Configure issue.yml playbook

```
[devops@ip-172-31-38-67 ansible]$ vim issue.yml
[devops@ip-172-31-38-67 ansible]$ cat issue.yml
---
- name: Configure /etc/issue for dev group
  hosts: dev
  become: true
  tasks:
    - name: Set /etc/issue for dev
      copy:
        dest: /etc/issue
        content: "development\n"

- name: Configure /etc/issue for test group
  hosts: test
  become: true
  tasks:
    - name: Set /etc/issue for test
      copy:
        dest: /etc/issue
        content: "test\n"
```

Step 2: Execute of issue.yml and verify /etc/issue on both hosts

```
[devops@ip-172-31-38-67 ansible]$ ansible-navigator run issue.yml -m stdout

PLAY [Configure /etc/issue for dev group] *****

TASK [Gathering Facts] *****
ok: [mumbai-client1]

TASK [Set /etc/issue for dev] *****
changed: [mumbai-client1]

PLAY [Configure /etc/issue for test group] *****

TASK [Gathering Facts] *****
ok: [mumbai-client2]

TASK [Set /etc/issue for test] *****
changed: [mumbai-client2]

PLAY RECAP *****
mumbai-client1      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
mumbai-client2      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
[devops@ip-172-31-38-67 ansible]$ ssh devops@52.66.203.197 'cat /etc/issue'
development
[devops@ip-172-31-38-67 ansible]$ ssh devops@13.127.215.162 'cat /etc/issue'
test
```

Task 6: Custom Apache Web Server Configuration

Objective: Configure Apache on dev with document root /webdev, content development, and symlink /webdev → /var/www/html.

- Installed httpd on dev group.
- Created index.html with content development.
- Created symbolic link /webdev → /var/www/html.
- Updated Apache DocumentRoot and Directory block; restarted httpd.

Step 1: Configure custom.yml playbook (Apache and webdev configuration)

```
[devops@ip-172-31-38-67 ansible]$ vim custom.yml
[devops@ip-172-31-38-67 ansible]$ cat custom.yml
---
- name: Configure custom web page on dev server
  hosts: dev
  become: true
  tasks:
    - name: Install Apache
      dnf:
        name: httpd
        state: present

    - name: Create web page with development content
      copy:
        dest: /var/www/html/index.html
        content: "<h1>development\n<h1>"

    - name: Create symbolic link /webdev to /var/www/html
      file:
        src: /var/www/html
        dest: /webdev
        state: link

    - name: Set Apache DocumentRoot to /webdev
      lineinfile:
        path: /etc/httpd/conf/httpd.conf
        regexp: '^DocumentRoot '
        line: 'DocumentRoot "/webdev"'

    - name: Update Apache Directory block for /webdev
      replace:
        path: /etc/httpd/conf/httpd.conf
        regexp: '<Directory "/var/www/html">'
        replace: '<Directory "/webdev">'

    - name: Start and enable Apache
      service:
        name: httpd
        state: restarted
        enabled: true
```

Step 2: Successful execution of custom.yml on mumbai-client1

```
[devops@ip-172-31-38-67 ansible]$ ansible-navigator run custom.yml -m stdout

PLAY [Configure custom web page on dev server] *****

TASK [Gathering Facts] *****
ok: [mumbai-client1]

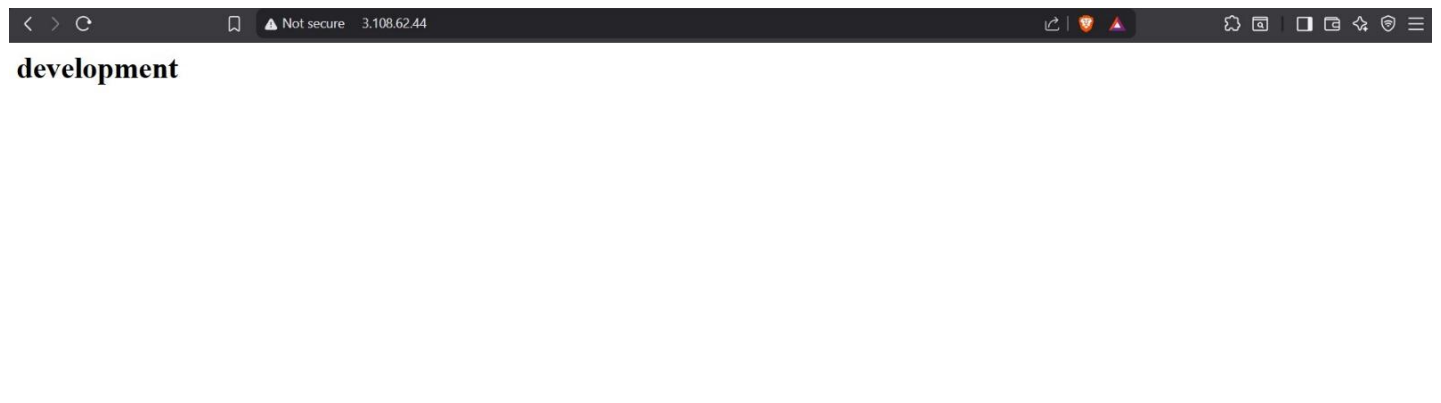
TASK [Install Apache] *****
ok: [mumbai-client1]

TASK [Start and enable Apache] *****
changed: [mumbai-client1]

TASK [Create custom dev homepage] *****
changed: [mumbai-client1]

PLAY RECAP *****
mumbai-client1      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Step 3: Browser verification — dev web page displaying development



Task 7: Git Integration

Objective: Push inventory, ansible.cfg, all playbooks, and roles/myrole to GitHub.

- Installed Git; initialized repository in /home/devops/ansible.
- Committed all project artifacts.
- Pushed the repository to GitHub.

Step 1: Install Git on the control node

```
[devops@ip-172-31-38-67 ansible]$ sudo dnf install -y git
Last metadata expiration check: 3:00:38 ago on Tue Jun 23 07:00:50 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
git	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux	53 k
Installing dependencies:				
git-core	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux	4.9 M
git-core-doc	noarch	2.50.1-1.amzn2023.0.1	amazonlinux	2.8 M
perl-Error	noarch	1:0.17030-2.amzn2023.0.1	amazonlinux	42 k
perl-File-Find	noarch	1.37-477.amzn2023.0.9	amazonlinux	26 k
perl-Git	noarch	2.50.1-1.amzn2023.0.1	amazonlinux	41 k
perl-TermReadKey	x86_64	2.38-9.amzn2023.0.3	amazonlinux	36 k
perl-lib	x86_64	0.65-477.amzn2023.0.9	amazonlinux	15 k

```
Transaction Summary
Install 8 Packages

Total download size: 7.9 M
Installed size: 41 M
Downloading Packages:
(1/8): git-2.50.1-1.amzn2023.0.1.x86_64.rpm           1.5 MB/s | 53 kB  00:00
(2/8): perl-Error-0.17030-2.amzn2023.0.1.noarch.rpm   1.5 MB/s | 42 kB  00:00
(3/8): git-core-doc-2.50.1-1.amzn2023.0.1.noarch.rpm  32 MB/s | 2.8 MB  00:00
(4/8): perl-File-Find-1.37-477.amzn2023.0.9.noarch.rpm 878 kB/s | 26 kB  00:00
(5/8): git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm     36 MB/s | 4.9 MB  00:00
(6/8): perl-Git-2.50.1-1.amzn2023.0.1.noarch.rpm      870 kB/s | 41 kB  00:00
(7/8): perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64.rpm 797 kB/s | 36 kB  00:00
(8/8): perl-lib-0.65-477.amzn2023.0.9.x86_64.rpm     652 kB/s | 15 kB  00:00
```

Step 2: Git repository initialization and user configuration

```
[devops@ip-172-31-38-67 ansible]$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/devops/ansible/.git/
[devops@ip-172-31-38-67 ansible]$ git branch -M main
[devops@ip-172-31-38-67 ansible]$ git config --global user.name "Adithyan Suresh Kumar"
[devops@ip-172-31-38-67 ansible]$ git config --global user.email "adithyansk2002@gmail.com"
```

Step 3: Stage project files with git add

```
[devops@ip-172-31-38-67 ansible]$ git add .
[devops@ip-172-31-38-67 ansible]$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .gitignore
    new file:   README.md
    new file:   ansible-navigator.log
    new file:   ansible-navigator.yml
    new file:   ansible.cfg
    new file:   custom-artifact-2026-06-23T09:53:33.224845+00:00.json
    new file:   custom.yml
    new file:   inventory
    new file:   issue-artifact-2026-06-23T09:33:08.717340+00:00.json
    new file:   issue-artifact-2026-06-23T09:41:03.777069+00:00.json
    new file:   issue.yml
    new file:   myrole-artifact-2026-06-23T09:19:30.428065+00:00.json
    new file:   myrole-artifact-2026-06-23T09:23:17.113802+00:00.json
    new file:   myrole.yml
    new file:   myrole/README.md
    new file:   myrole/defaults/main.yml
    new file:   myrole/handlers/main.yml
    new file:   myrole/meta/main.yml
    new file:   myrole/tasks/main.yml
    new file:   myrole/tests/inventory
    new file:   myrole/tests/test.yml
    new file:   myrole/vars/main.yml
    new file:   packages-artifact-2026-06-23T08:30:47.910848+00:00.json
    new file:   packages-artifact-2026-06-23T08:41:30.334199+00:00.json
    new file:   packages-artifact-2026-06-23T08:46:30.961644+00:00.json
    new file:   packages-artifact-2026-06-23T08:56:33.201850+00:00.json
```

Step 4: Initial commit to local repository

```
[devops@ip-172-31-38-67 ansible]$ git commit -m "Initial Commit"
[main (root-commit) 46e6d57] Initial Commit
27 files changed, 16208 insertions(+)
create mode 100644 .gitignore
create mode 100644 README.md
create mode 100644 ansible-navigator.log
create mode 100644 ansible-navigator.yml
create mode 100644 ansible.cfg
create mode 100644 custom-artifact-2026-06-23T09:53:33.224845+00:00.json
create mode 100644 custom.yml
create mode 100644 inventory
create mode 100644 issue-artifact-2026-06-23T09:33:08.717340+00:00.json
create mode 100644 issue-artifact-2026-06-23T09:41:03.777069+00:00.json
create mode 100644 issue.yml
create mode 100644 myrole-artifact-2026-06-23T09:19:30.428065+00:00.json
create mode 100644 myrole-artifact-2026-06-23T09:23:17.113802+00:00.json
create mode 100644 myrole.yml
create mode 100644 myrole/README.md
create mode 100644 myrole/defaults/main.yml
create mode 100644 myrole/handlers/main.yml
create mode 100644 myrole/meta/main.yml
create mode 100644 myrole/tasks/main.yml
create mode 100644 myrole/tests/inventory
create mode 100644 myrole/tests/test.yml
create mode 100644 myrole/vars/main.yml
create mode 100644 packages-artifact-2026-06-23T08:30:47.910848+00:00.json
create mode 100644 packages-artifact-2026-06-23T08:41:30.334199+00:00.json
create mode 100644 packages-artifact-2026-06-23T08:46:30.961644+00:00.json
create mode 100644 packages-artifact-2026-06-23T08:56:33.201850+00:00.json
create mode 100644 packages.yml
```

Step 5: Push repository to GitHub

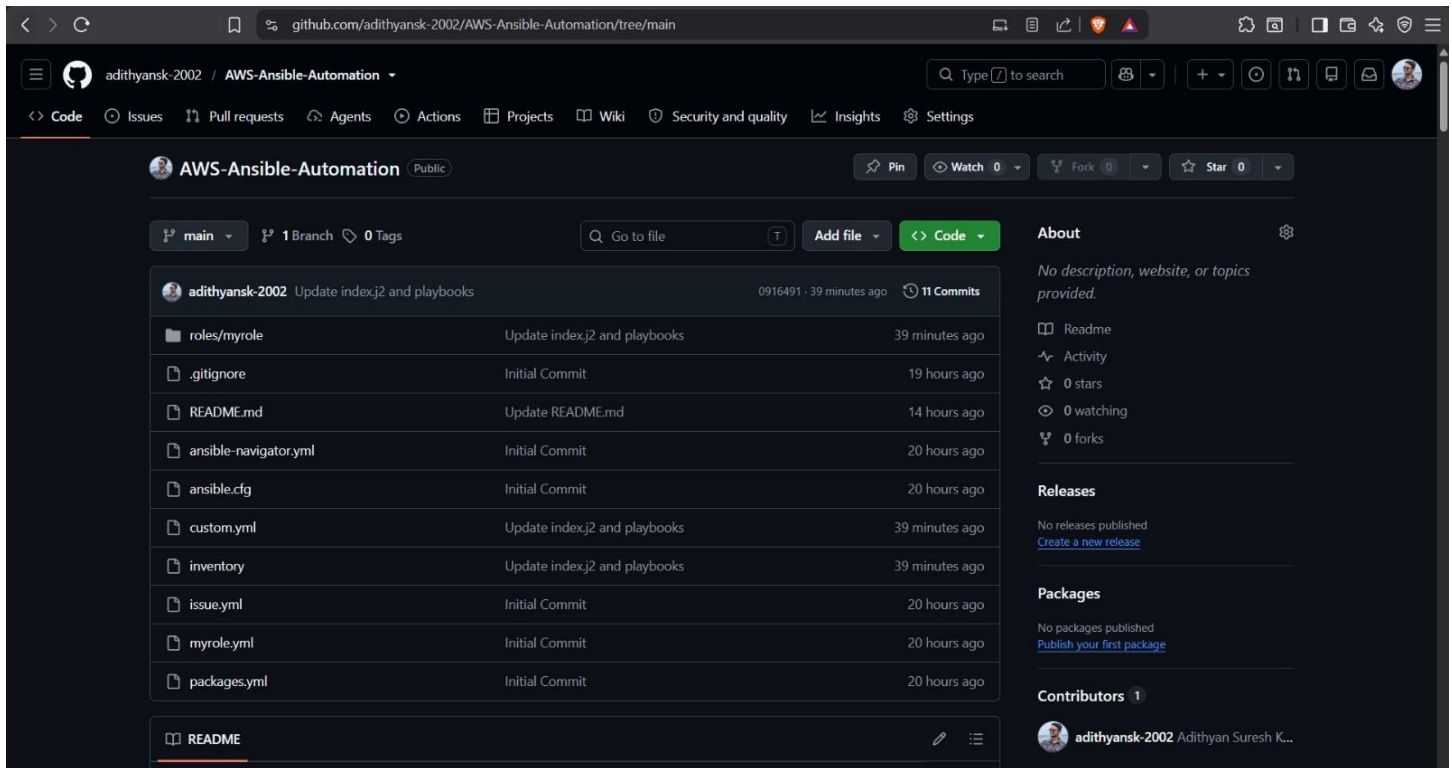
```
[devops@ip-172-31-38-67 ansible]$ git remote add origin https://github.com/adithyansk-2002/AWS-Ansible-Automation
[devops@ip-172-31-38-67 ansible]$ git push -u origin main -f
Username for 'https://github.com': adithyansk-2002
Password for 'https://adithyansk-2002@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/adithyansk-2002/AWS-Ansible-Automation/'
[devops@ip-172-31-38-67 ansible]$ git push -u origin main -f
Username for 'https://github.com': adithyansk-2002
Password for 'https://adithyansk-2002@github.com':
Enumerating objects: 36, done.
Counting objects: 100% (36/36), done.
Delta compression using up to 2 threads
Compressing objects: 100% (25/25), done.
Writing objects: 100% (36/36), 37.87 KiB | 2.52 MiB/s, done.
Total 36 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), done.
To https://github.com/adithyansk-2002/AWS-Ansible-Automation
+ ed1451c...46e6d57 main -> main (forced update)
branch 'main' set up to track 'origin/main'.
```

Step 6: Git status, log, and remote verification

```
[devops@ip-172-31-38-67 ansible]$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
[devops@ip-172-31-38-67 ansible]$ git log --oneline
46e6d57 (HEAD -> main, origin/main) Initial Commit
[devops@ip-172-31-38-67 ansible]$ git remote -v
origin https://github.com/adithyansk-2002/AWS-Ansible-Automation (fetch)
origin https://github.com/adithyansk-2002/AWS-Ansible-Automation (push)
```

Step 7: GitHub repository page — AWS-Ansible-Automation



After completing Mission 1, the GitHub repository was used to recreate the same Ansible environment in the Hyderabad region, demonstrating the portability and reusability of the automation setup.

4. Mission 2 — Hyderabad (ap-south-2)

Instances: hyderabad-control, hyderabad-client1 (dev), hyderabad-client2 (test)

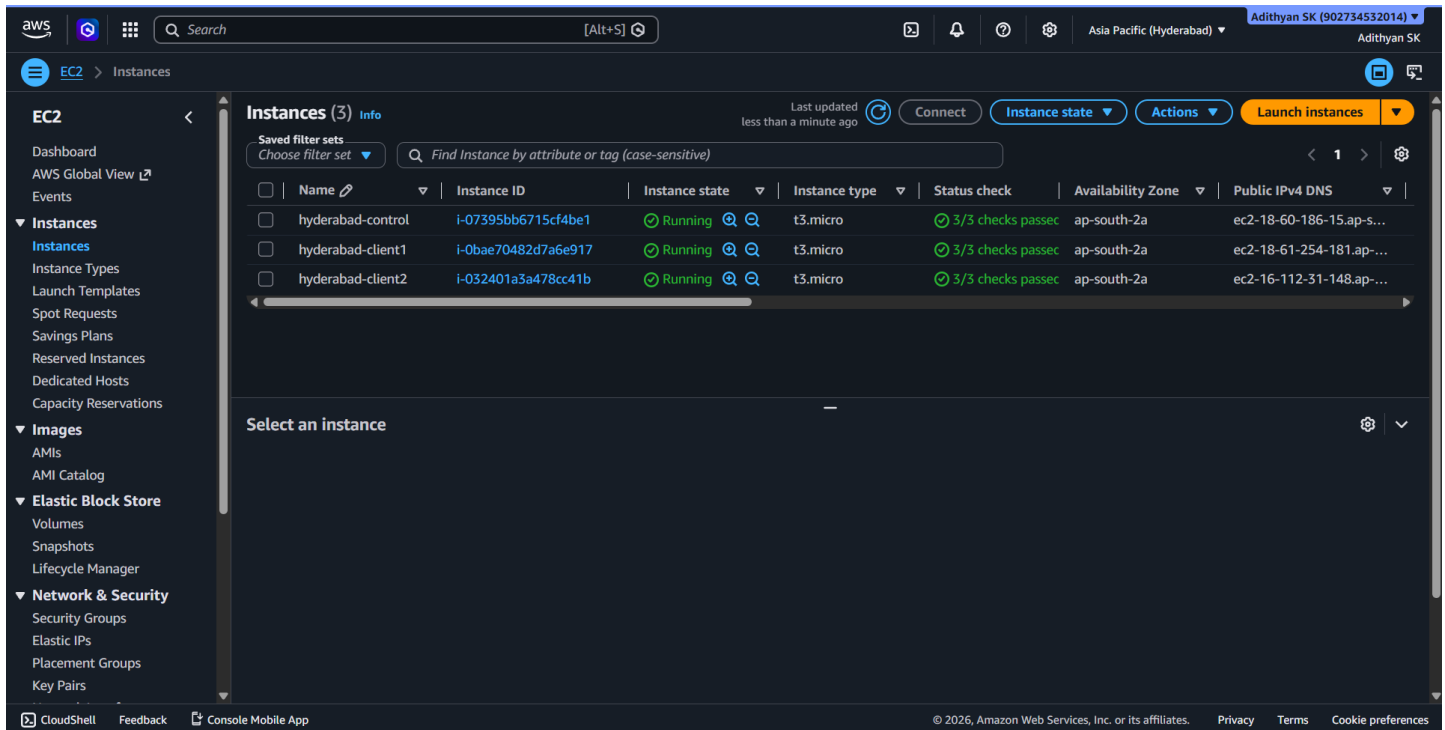
User: clone

Path: /home/clone/ansible

Task 1: Create New Environment

- Launched three t3.micro instances in ap-south-2a.
- Created clone user on all machines.
- Manually shared SSH keys (same method as Mission 1).
- Verified SSH from hyderabad-control to both clients.

Step 1: Run 3 instances in EC2 Console in Hyderabad (hyderabad-control, hyderabad-client1, hyderabad-client2)



Step 2: Verify SSH connectivity from control node to hyderabad-client1 and hyderabad-client2 after sharing public keys to both instances

```
[clone@ip-172-31-12-215 ~]$ ssh clone@18.60.39.33
#_
~\_##### Amazon Linux 2023
~~\_#####\
~~\#####|
~~\##/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~' '->
~~~~
~~~.~.~
~~\_/ \_/ \_/
~~/m/' \_/

Last login: Wed Jun 24 06:43:40 2026 from 16.112.67.244
[clone@ip-172-31-7-181 ~]$ exit
logout
Connection to 18.60.39.33 closed.
[clone@ip-172-31-12-215 ~]$ ssh clone@16.112.70.141
#_
~\_##### Amazon Linux 2023
~~\_#####\
~~\#####|
~~\##/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~' '->
~~~~
~~~.~.~
~~\_/ \_/ \_/
~~/m/' \_/

Last login: Wed Jun 24 06:43:55 2026 from 16.112.67.244
[clone@ip-172-31-5-39 ~]$ exit
logout
Connection to 16.112.70.141 closed.
```

Task 2: Clone Repository and Rebuild Environment

- Installed Docker, Ansible, and ansible-navigator on hyderabad-control.
- Cloned Mission 1 repository to /home/clone/ansible.
- Updated inventory with Hyderabad IP addresses and clone user.
- Updated ansible.cfg and ansible-navigator.yml paths.
- Verified ansible all -m ping.
- Re-executed packages.yml, myrole.yml, issue.yml, and custom.yml.

Step 1: Install Docker on hyderabad-control

```
[clone@ip-172-31-12-215 ~]$ sudo dnf install -y docker
Last metadata expiration check: 0:00:14 ago on Wed Jun 24 06:52:02 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
docker	x86_64	25.0.16-1.amzn2023.0.2	amazonlinux	46 M
Installing dependencies:				
container-selinux	noarch	4:2.245.0-1.amzn2023	amazonlinux	58 k
containerd	x86_64	2.2.4-1.amzn2023.0.3	amazonlinux	24 M
iptables-libs	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	401 k
iptables-nft	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	183 k
libcgroup	x86_64	3.0-1.amzn2023.0.1	amazonlinux	75 k
libnetfilter_conntrack	x86_64	1.0.8-2.amzn2023.0.2	amazonlinux	58 k
libnftnl	x86_64	1.0.1-19.amzn2023.0.2	amazonlinux	30 k
libnftnl	x86_64	1.2.2-2.amzn2023.0.2	amazonlinux	84 k
pigz	x86_64	2.5-1.amzn2023.0.4	amazonlinux	83 k
runC	x86_64	1.3.5-1.amzn2023.0.2	amazonlinux	3.9 M

```
Transaction Summary
Install 11 Packages

Total download size: 76 M
Installed size: 284 M
Downloading Packages:
(1/11): container-selinux-2.245.0-1.amzn2023.noarch.rpm                1.5 MB/s | 58 kB    00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm                12 MB/s | 401 kB   00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                 5.6 MB/s | 183 kB   00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm                     3.4 MB/s | 75 kB    00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm       2.0 MB/s | 58 kB    00:00
(6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm                   1.0 MB/s | 30 kB    00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                     2.5 MB/s | 84 kB    00:00
(8/11): pigz-2.5-1.amzn2023.0.4.x86_64.rpm                           3.2 MB/s | 83 kB    00:00
```

Step 2: Docker service enabled and running on hyderabad-control

```
[clone@ip-172-31-12-215 ~]$ sudo systemctl enable --now docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[clone@ip-172-31-12-215 ~]$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-06-24 06:53:15 UTC; 55s ago
 TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Process: 28795 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
   Process: 28796 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Main PID: 28797 (dockerd)
    Tasks: 9
   Memory: 30.5M
    CPU: 335ms
   CGroup: /system.slice/docker.service
           └─28797 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Jun 24 06:53:15 ip-172-31-12-215.ap-south-2.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Jun 24 06:53:15 ip-172-31-12-215.ap-south-2.compute.internal dockerd[28797]: time="2026-06-24T06:53:15.373238591Z" level=info msg="Starting up"
Jun 24 06:53:15 ip-172-31-12-215.ap-south-2.compute.internal dockerd[28797]: time="2026-06-24T06:53:15.445107553Z" level=info msg="Loading containers: start."
Jun 24 06:53:15 ip-172-31-12-215.ap-south-2.compute.internal dockerd[28797]: time="2026-06-24T06:53:15.864018374Z" level=info msg="Loading containers: done."
Jun 24 06:53:15 ip-172-31-12-215.ap-south-2.compute.internal dockerd[28797]: time="2026-06-24T06:53:15.902166854Z" level=info msg="Docker daemon" commit=6fdf0a..n=25.0.16
Jun 24 06:53:15 ip-172-31-12-215.ap-south-2.compute.internal dockerd[28797]: time="2026-06-24T06:53:15.902350813Z" level=info msg="Daemon has completed initialization"
Jun 24 06:53:15 ip-172-31-12-215.ap-south-2.compute.internal dockerd[28797]: time="2026-06-24T06:53:15.929440896Z" level=info msg="API listen on /run/docker.sock"
Jun 24 06:53:15 ip-172-31-12-215.ap-south-2.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
```

Step 3: Install Python 3, pip, and Git

```
[clone@ip-172-31-12-215 ~]$ sudo dnf install -y python3 python3-pip git
Last metadata expiration check: 0:04:15 ago on Wed Jun 24 06:52:02 2026.
Package python3-3.9.25-1.amzn2023.0.6.x86_64 is already installed.
Dependencies resolved.

=====
```

Package	Architecture	Version	Repository	Size
Installing:				
git	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux	53 k
python3-pip	noarch	21.3.1-2.amzn2023.0.20	amazonlinux	1.8 M
Installing dependencies:				
git-core	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux	4.9 M
git-core-doc	noarch	2.50.1-1.amzn2023.0.1	amazonlinux	2.8 M
perl-Error	noarch	1:0.17030-2.amzn2023.0.1	amazonlinux	42 k
perl-File-Find	noarch	1.37-477.amzn2023.0.9	amazonlinux	26 k
perl-Git	noarch	2.50.1-1.amzn2023.0.1	amazonlinux	41 k
perl-TermReadKey	x86_64	2.38-9.amzn2023.0.3	amazonlinux	36 k
perl-lib	x86_64	0.65-477.amzn2023.0.9	amazonlinux	15 k
Installing weak dependencies:				
libxcrypt-compat	x86_64	4.4.33-7.amzn2023	amazonlinux	92 k

```
=====
Transaction Summary
-----
Install 10 Packages

Total download size: 9.8 M
Installed size: 53 M
Downloading Packages:
(1/10): git-2.50.1-1.amzn2023.0.1.x86_64.rpm                1.3 MB/s | 53 kB    00:00
(2/10): libxcrypt-compat-4.4.33-7.amzn2023.x86_64.rpm       2.7 MB/s | 92 kB    00:00
(3/10): git-core-doc-2.50.1-1.amzn2023.0.1.noarch.rpm       32 MB/s | 2.8 MB    00:00
(4/10): perl-Error-0.17030-2.amzn2023.0.1.noarch.rpm        1.5 MB/s | 42 kB    00:00
(5/10): perl-File-Find-1.37-477.amzn2023.0.9.noarch.rpm     1.0 MB/s | 26 kB    00:00
(6/10): git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm          33 MB/s | 4.9 MB    00:00
(7/10): perl-Git-2.50.1-1.amzn2023.0.1.noarch.rpm           808 kB/s | 41 kB    00:00
```

Step 4: Install ansible-navigator, ansible-builder, and ansible

```
[clone@ip-172-31-12-215 ~]$ python3 -m pip install --user ansible-navigator ansible-builder ansible
Collecting ansible-navigator
  Downloading ansible_navigator-24.2.0-py3-none-any.whl (297 kB)
    |#####| 297 kB 16.4 MB/s
Collecting ansible-builder
  Downloading ansible_builder-3.1.1-py3-none-any.whl (46 kB)
    |#####| 46 kB 7.5 MB/s
Collecting ansible
  Downloading ansible-8.7.0-py3-none-any.whl (48.4 MB)
    |#####| 48.4 MB 35.3 MB/s
Requirement already satisfied: Jinja2 in /usr/lib/python3.9/site-packages (from ansible-navigator) (2.11.3)
Collecting ansible-lint>=6.19.0
  Downloading ansible_lint-6.22.2-py3-none-any.whl (297 kB)
    |#####| 297 kB 60.0 MB/s
Requirement already satisfied: jsonschema in /usr/lib/python3.9/site-packages (from ansible-navigator) (3.2.0)
Collecting ansible-runner<3,>=2.3.1
  Downloading ansible_runner-2.4.3-py3-none-any.whl (80 kB)
    |#####| 80 kB 18.6 MB/s
Collecting importlib-metadata
  Downloading importlib_metadata-8.7.1-py3-none-any.whl (27 kB)
Collecting tzdata
  Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
    |#####| 349 kB 62.5 MB/s
Collecting onigurumacffi<2,>=1.1.0
  Downloading onigurumacffi-1.4.1-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (603 kB)
    |#####| 603 kB 57.2 MB/s
Requirement already satisfied: pyyaml in /usr/lib64/python3.9/site-packages (from ansible-navigator) (5.4.1)
Collecting packaging
  Downloading packaging-26.2-py3-none-any.whl (100 kB)
    |#####| 100 kB 14.3 MB/s
Collecting bindep
  Downloading bindep-2.14.0-py3-none-any.whl (35 kB)
Collecting ansible-core~2.15.7
  Downloading ansible_core-2.15.13-py3-none-any.whl (2.3 MB)
    |#####| 2.3 MB 54.4 MB/s
```

Step 5: Tool version verification on hyderabad-control

```
[clone@ip-172-31-12-215 ~]$ source ~/.bashrc
[clone@ip-172-31-12-215 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[clone@ip-172-31-12-215 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[clone@ip-172-31-12-215 ~]$ ansible-navigator --version
ansible-navigator 24.2.0
[clone@ip-172-31-12-215 ~]$ ansible --version
ansible [core 2.15.13]
  config file = None
  configured module search path = ['/home/clone/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /home/clone/.local/lib/python3.9/site-packages/ansible
  ansible collection location = /home/clone/.ansible/collections:/usr/share/ansible/collections
  executable location = /home/clone/.local/bin/ansible
  python version = 3.9.25 (main, May 25 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
[clone@ip-172-31-12-215 ~]$ git --version
git version 2.50.1
```

Step 6: Clone GitHub repository into /home/clone/ansible

```
[clone@ip-172-31-12-215 ~]$ git clone https://github.com/adithyansk-2002/AWS-Ansible-Automation ansible
Cloning into 'ansible'...
remote: Enumerating objects: 79, done.
remote: Counting objects: 100% (79/79), done.
remote: Compressing objects: 100% (41/41), done.
remote: Total 79 (delta 36), reused 64 (delta 21), pack-reused 0 (from 0)
Receiving objects: 100% (79/79), 48.20 KiB | 2.68 MiB/s, done.
Resolving deltas: 100% (36/36), done.
[clone@ip-172-31-12-215 ~]$
[clone@ip-172-31-12-215 ~]$
[clone@ip-172-31-12-215 ~]$ cd /home/clone/ansible
[clone@ip-172-31-12-215 ansible]$ ls
README.md  ansible-navigator.yml  ansible.cfg  custom.yml  inventory  issue.yml  myrole.yml  packages.yml  roles
```

Step 7: Hyderabad inventory configuration with updated IP addresses

```
[clone@ip-172-31-12-215 ansible]$ vim inventory
[clone@ip-172-31-12-215 ansible]$ cat inventory
[dev]
hyderabad-client1 ansible_host=18.60.39.33

[test]
hyderabad-client2 ansible_host=16.112.70.141

[all:vars]
ansible_user=clone
ansible_python_interpreter=/usr/bin/python3
```

Step 8: Update ansible.cfg for clone user and Hyderabad paths

```
[clone@ip-172-31-12-215 ansible]$ vim ansible.cfg
[clone@ip-172-31-12-215 ansible]$ cat ansible.cfg
[defaults]
inventory = /home/clone/ansible/inventory
host_key_checking = False
retry_files_enabled = False
remote user = clone
```

Step 9: Update ansible-navigator.yml for Hyderabad environment

```
[clone@ip-172-31-12-215 ansible]$ vim ansible-navigator.yml
[clone@ip-172-31-12-215 ansible]$ cat ansible-navigator.yml
---
ansible-navigator:
  execution-environment:
    enabled: true
    image: ghcr.io/ansible/creator-ee:v0.22.0
    pull:
      policy: missing
  ansible:
    config:
      path: /home/clone/ansible/ansible.cfg
```

Step 10: Successful ansible all -m ping on Hyderabad clients

```
[clone@ip-172-31-12-215 ansible]$ ansible all -m ping
hyderabad-client2 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
hyderabad-client1 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

Step 11: Successful execution of packages.yml in Hyderabad

Play name	OK	Changed	Unreachable	Failed	Skipped	Ignored	In progress	Task count	Progress
0 Install common packages on dev and test	4	2	0	0	0	0	0	4	Complete
1 Update all packages and install Development Tools on dev	3	1	0	0	0	0	0	3	Complete

^b/PgUp page up ^f/PgDn page down | scroll esc back [0-9] goto :help help Successful

Step 12: Successful execution of myrole.yml on hyderabad-client2

```
[clone@ip-172-31-12-215 ansible]$ ansible-navigator run myrole.yml -m stdout

PLAY [Apply myrole to test group] *****

TASK [Gathering Facts] *****
ok: [hyderabad-client2]

TASK [myrole : Install Apache web server] *****
ok: [hyderabad-client2]

TASK [myrole : Deploy index page from template] *****
changed: [hyderabad-client2]

TASK [myrole : Start and enable Apache] *****
changed: [hyderabad-client2]

PLAY RECAP *****
hyderabad-client2      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Step 13: Browser verification — Hyderabad test web page (hostname and IP)

Not secure 16.112.70.141

Welcome to ip-172-31-5-39.ap-south-2.compute.internal on 172.31.5.39

Step 14: Successful execution of issue.yml in Hyderabad

Play_name	Ok	Changed	Unreachable	Failed	Skipped	Ignored	In progress	Task count	Progress
0 Configure /etc/issue for dev group	2	1	0	0	0	0	0	2	Complete
1 Configure /etc/issue for test group	2	1	0	0	0	0	0	2	Complete

^b/PgUp page up ^f/PgDn page down ↑↓ scroll esc back [0-9] goto :help help Successful

Step 15: Verification of /etc/issue — development (dev) and test (test)

```
[clone@ip-172-31-12-215 ansible]$ ssh clone@18.60.39.33 'cat /etc/issue'
development
[clone@ip-172-31-12-215 ansible]$ ssh clone@16.112.70.141 'cat /etc/issue'
test
```

Step 16: Successful execution of custom.yml in Hyderabad

Play_name	Ok	Changed	Unreachable	Failed	Skipped	Ignored	In_progress	Task_count	Progress
0 Configure custom web page on dev server	7	5	0	0	0	0	0	7	Complete

^b/PgUp page up
^f/PgDn page down
↑ scroll
esc back
[0-9] goto
:help help
Successful

Step 17: Browser verification — Hyderabad dev web page displaying development

development

5. Conclusion

Both Mission 1 and Mission 2 were completed successfully. A fully automated Ansible environment was built on AWS EC2 in Mumbai, version-controlled on GitHub, and reproduced in Hyderabad by cloning the repository and updating inventory and configuration paths. All playbooks were executed exclusively through ansible-navigator inside a Docker execution environment, meeting all assignment requirements.